Halward Inspection Services Ltd.

Inspection Report

Site: Hamilton Plant

No.	Status	Date
HIS-2010-0251	Issued	2010-06-21

UT Thickness Report - 4-RA-104 CUI Locations

Scope and Responsibilities

This report details the ultrasonic thickness examination of line 4-RA-104 within the ATU-100 at Hamilton Plant. The scope was limited to specific areas where insulation was stripped following indications of external wall loss. Kevin Doucette, acting as the lead UT technician, was responsible for the execution of the NDE and the verification of data against site retirement limits.

The inspection was performed under QSCP-NDT-001 Rev 1, which mandates the use of gauges with waveform displays and requires scanning at elbows and tees rather than spot measurements. All activities were conducted in accordance with the requirements for piping inspection and damage mechanism identification established in API 570 and API RP 571.

Background and Trigger

The requirement for this targeted UT examination originated from HIS-2010-0233. That profile radiography report identified wall loss at two locations on the insulated 4-RA-104, while 4-LA-104 showed only superficial external pitting of 0.5 mm or less. Consequently, insulation was removed from the suspect spools on 4-RA-104 to allow for direct surface access and precise thickness measurement.

Line 4-RA-104 is an A106 Gr.B rich-side exchanger line with an OD of 114.3 mm and a nominal thickness of 8.56 mm. The design limits for this circuit are 862 kPa(g) and 127 °C. Per site rules, the retirement thickness t-min is 3.81 mm, and any reading below this value requires the issuance of an NCR.

Inspection Results

Ultrasonic scanning was performed at the stripped locations to characterize the extent of Corrosion Under Insulation. At CUI location 1, which carries CMLs -03 to -05, the minimum thickness recorded was 3.64 mm. At CUI location 2, also carrying CMLs -03 to -05, the minimum thickness recorded was 3.72 mm.

Both recorded values are below the established t-min of 3.81 mm. These findings confirm the external CUI wall loss initially indicated by radiography. Because these readings have reached the retirement threshold, they are raised as an NCR in accordance with Hamilton Plant asset integrity requirements.

Remaining Life and Scheduling

Remaining life calculations must utilize the larger of the long-term and short-term corrosion rates. Given that the current readings of 3.64 mm and 3.72 mm are below the t-min of 3.81 mm, the remaining life is zero. This necessitates immediate engineering assessment for repair or replacement of the affected spools.

Under normal circumstances for Class 2 amine service, the maximum thickness interval is 10 years. However, the current condition of 4-RA-104 supersedes standard intervals. Kevin Doucette has documented these findings to facilitate the necessary corrective actions per API 570 protocols.

Sign-off	Name	Function	Date	Signature
Prepared	Kevin Doucette	lead UT technician, SNT-TC-1A UT Level II	2010-06-21	_____